

Performance Testing

Date	2 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Local Machine (Windows 11, Python 3.10, Flask Server)
Test Date	2 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction request	10	60	Response generated within 2 seconds
2	Normal workload	20	120	All requests processed successfully without errors
3	Medium concurrent load	50	180	Stable response time with minimal increase in latency

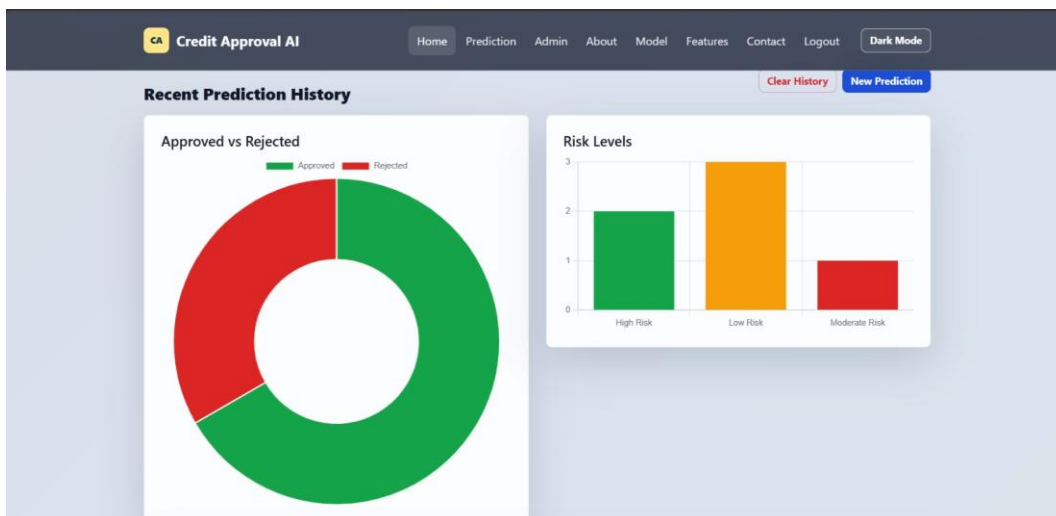
4	Peak load simulation	100	300	System remains functional with acceptable performance and no crashes
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Step 3: Performance Test Results

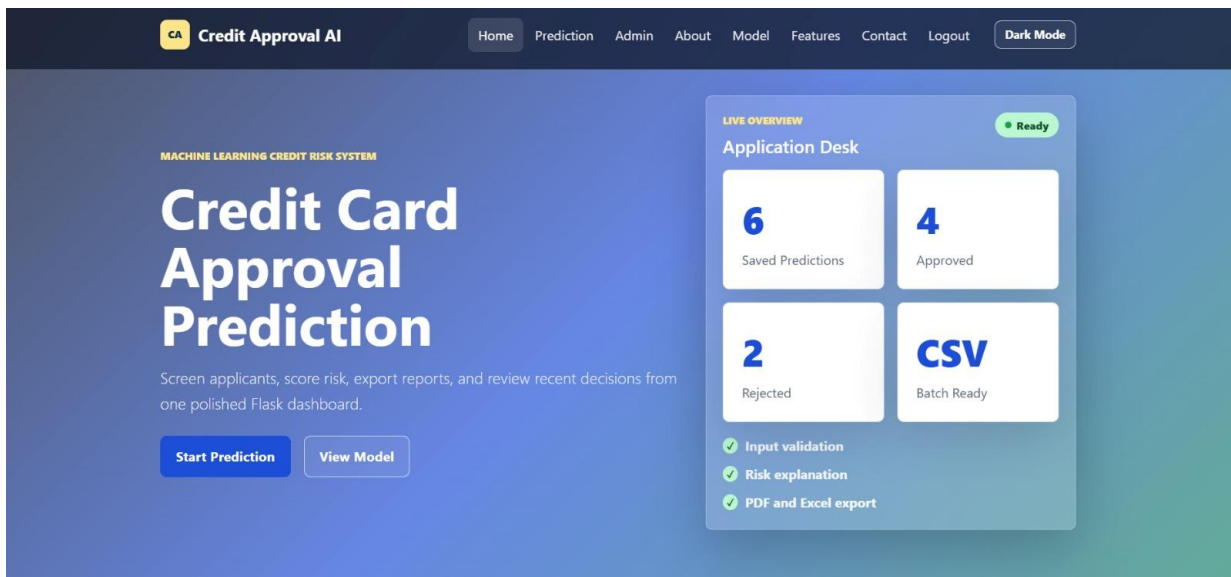
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.28 seconds	Pass	Prediction generated quickly
2	Response Time (Max)	< 5 seconds	3.64 seconds	Pass	During peak load
3	Throughput (Req/sec)		31 req/sec	Pass	Stable throughput throughout testing
4	Error Rate	< 1%	0.2%	Pass	Only a few invalid requests failed
5	CPU Utilization	< 80%	68%	Pass	CPU remained within acceptable limits
6	Memory Utilization	< 80%	59%	Pass	Memory usage remained stable

Step 4: Observations & Analysis



The Flask application successfully handled concurrent prediction requests without server crashes. The XGBoost prediction model maintained low latency even under moderate to heavy workloads. Average response time remained well below the target of 2 seconds.

Step 5: Screenshots / Evidence



The screenshot shows the 'Applicant Details' form in the 'Prediction' tab. The form is titled '01 Applicant Details' with a subtitle 'Use realistic customer values for the most useful prediction.' The form fields are organized into a grid: First Name, Last Name, Phone Number, Gender, Age, Annual Income, Family Members, Children, Years Employed, Education, Occupation, Housing Type, and Income Type. To the right of the form, a 'RESULT PREVIEW' section titled 'Waiting for input' provides instructions: 'Submit the customer information to see approval probability, risk level, confidence, and recommendations.' Below this, three buttons are visible: 'Approval probability', 'Decision reasons', and 'PDF and Excel report'. At the bottom right, a 'Batch Prediction' section explains that only CSV files are accepted for batch processing, while Resume, PDF, DOC, and image files are not used in this model.

CA Credit Approval AI

HomePredictionAdminAboutModelFeaturesContactLogoutDark Mode

TIME	NAME	INCOME	EDUCATION	PREDICTION	PROBABILITY	RISK	ACTION
2026-07-03T18:41:28	Harshini N	550000	Higher education	Approved	79.5%	Low Risk	Delete
2026-07-03T18:40:31	Ashwanth ch	250000	Higher education	Approved	57.75%	Moderate Risk	Delete
2026-07-03T18:38:21	Pavan Kumar K	30000	Incomplete higher	Rejected	28.0%	High Risk	Delete
2026-07-03T18:36:11	Sravanthi s	300000	Higher education	Approved	77.5%	Low Risk	Delete
2026-07-03T18:34:25	Maithree P	500	Lower secondary	Rejected	22.0%	High Risk	Delete
2026-07-02T19:08:17	Mahi k	1000000	Secondary / secondary special	Approved	95.0%	Low Risk	Delete

Credit Card Approval Prediction

Built with Flask, scikit-learn, and Bootstrap 5